

EXERCISE 2: MEMORY MATCHING FLUTTER GAME

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1. Project Description

This project is a memory matching game developed using Flutter and Dart. The player needs to find all matching animal card pairs. The game contains 20 cards, including 10 different animal pairs. When the player taps a card, the card flips and shows the animal image. If two selected cards match, they remain open with a check mark. If they do not match, they flip back automatically after a short delay.

2. Main Features

- Built with Flutter and Dart
- 20 memory cards with 10 matching animal pairs
- Random card order every time the game starts
- Tap cards to reveal animal images
- Automatic matching logic
- Non-matching cards flip back after a short delay
- Move counter
- Restart button
- Green check mark for matched cards
- Congratulations dialog after all pairs are matched
- Responsive design for Chrome web and Android emulator

3. Project Structure

memory_match_app/

 android/

 assets/

 images/

 bear.png

 cat.png

```
dog.png
fox.png
lion.png
monkey.png
panda.png
rabbit.png
tiger.png
zebra.png

lib/
  main.dart

test/

web/

windows/

pubspec.yaml

README.pdf
```

4. How to Configure the Project

Step 1: Open Android Studio.

Step 2: Create a new Flutter project named memory_match_app.

Step 3: Create an assets/images folder in the project root.

Step 4: Add 10 animal image files into assets/images.

Step 5: Register the image folder in pubspec.yaml:

```
flutter:

  uses-material-design: true

  assets:
    - assets/images/
```

Step 6: Run flutter pub get to load the assets and dependencies.

5. How to Run the App

To run on Chrome web:

- Step 1: Open the project in Android Studio.
- Step 2: Select Chrome (web) from the device list.
- Step 3: Click the green Run button.
- Step 4: The app will open in the browser.

To run on Android Emulator:

- Step 1: Open Device Manager in Android Studio.
- Step 2: Start an Android emulator.
- Step 3: Select the emulator from the device list.
- Step 4: Click the green Run button.

6. Implementation Explanation

The app uses a StatefulWidget to manage the game state. Each card has an id, image path, emoji fallback, face-up status, and matched status.

At the beginning of the game, the app creates two copies of each animal card to form a matching pair. Then, all cards are shuffled randomly.

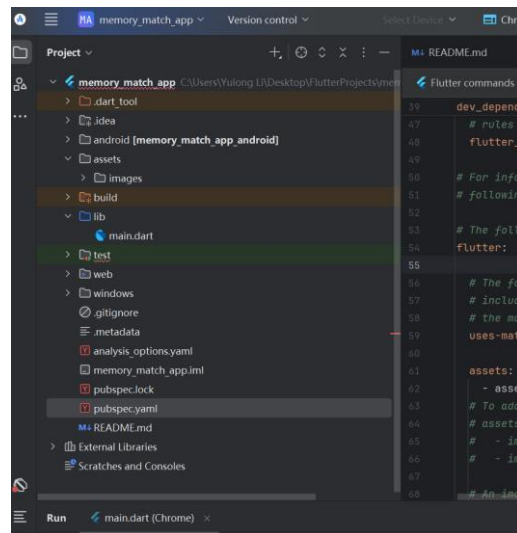
When the user taps a card, the card is flipped and the animal image is shown. If it is the first selected card, the app waits for the second selected card. When the second card is selected, the app compares the two card ids.

If the two ids are the same, the two cards are marked as matched and remain open with a green check mark. If the two ids are different, the app waits for a short time and then flips both cards back.

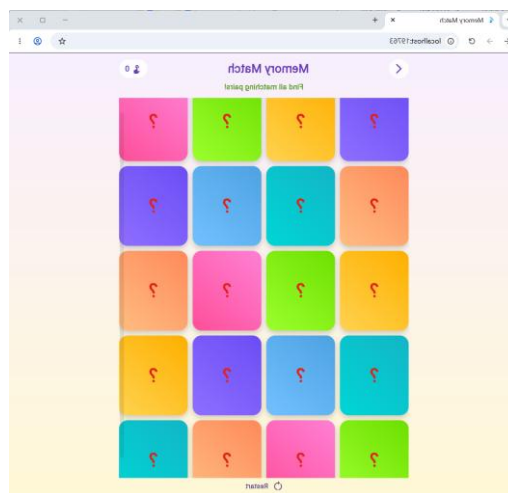
When all 10 pairs are matched, a Congratulations dialog is displayed.

7. Screenshots

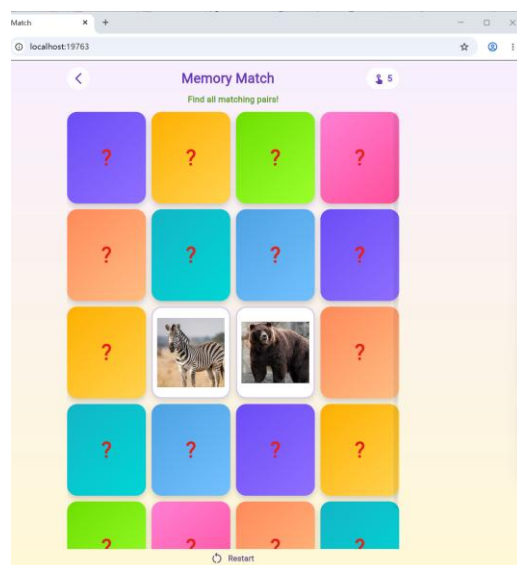
Screenshot 1: Project Structure



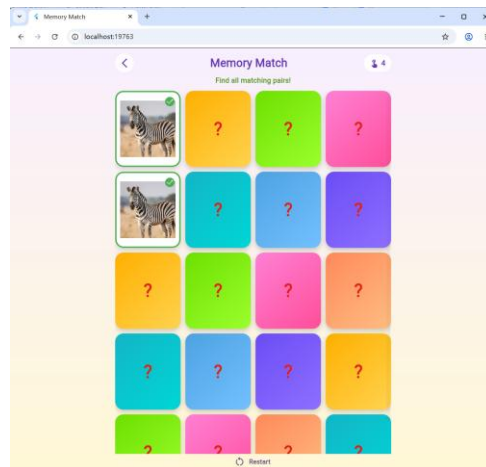
Screenshot 2: Initial Game Screen



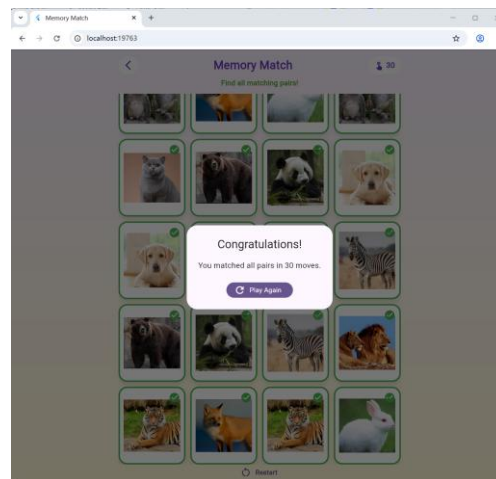
Screenshot 3: Cards Flipped



Screenshot 4: Matched Cards



Screenshot 5: Winning Dialog



8. Conclusion

This Flutter/Dart project demonstrates basic mobile app development, image assets, user interaction, state management, conditional rendering, timer-based game logic, and responsive UI design.